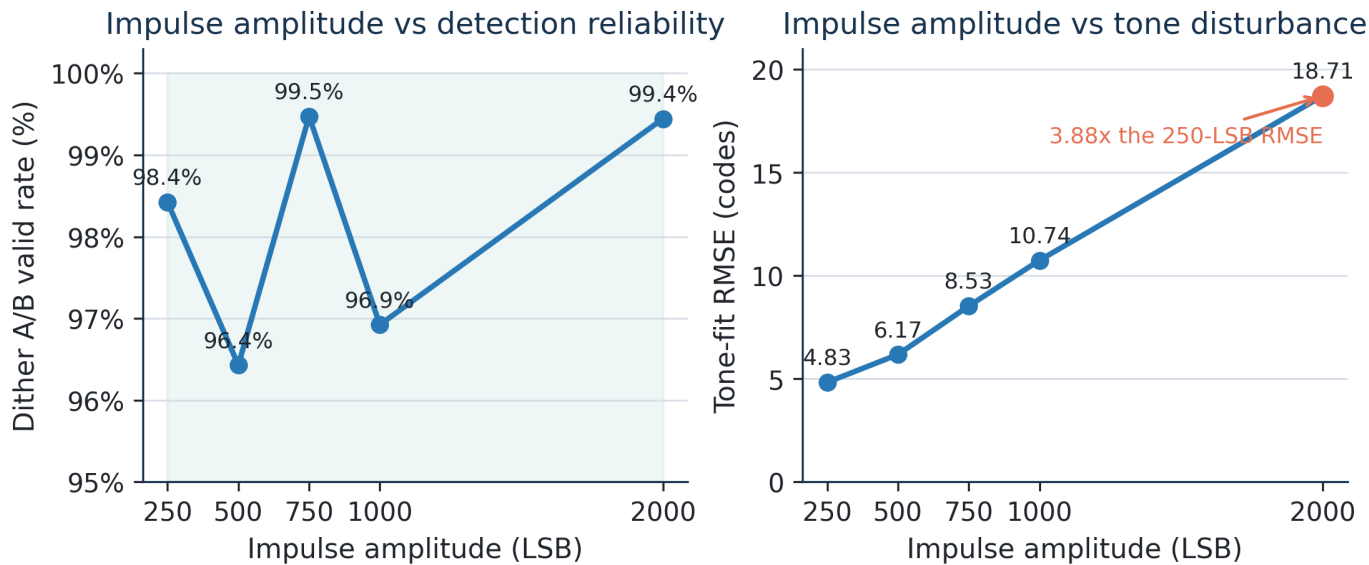


TIADC Impulse Dither - Board-Level Operating-Region Update

Research update for Professor Liu | 27 August 2026

Key takeaway. Board sweeps show that sparse, low-amplitude impulse excitation remains reliably detectable while introducing limited tone-path disturbance. Very narrow pulses are viable; excessively sparse repetition reduces detection. This strengthens the practical injection operating-region case, while the known gain and fine-skew limitations remain unchanged.

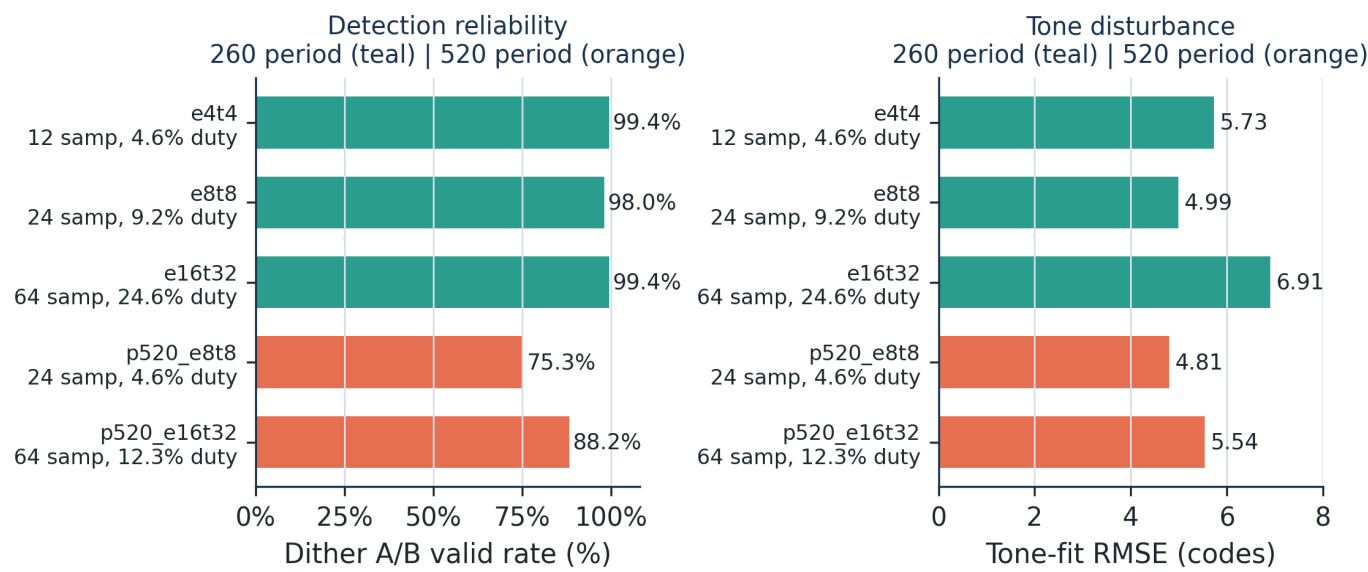
1. Board-Level Amplitude Sweep



Across 250-2000 LSB, A/B event validity is 96.4-99.5% and timing correlation is 0.99881-0.99993. Detection therefore remains strong across the tested range. The cost of stronger injection is visible in the tone fit: RMSE rises from 4.83 codes at 250 LSB to 18.71 codes at 2000 LSB (+13.89 codes; 3.88x). The sweep identifies a broad low-disturbance operating region, not a single globally optimal amplitude; larger impulses are not inherently better.

Figure 1. Values are means from the committed board timing/skew captures. The shaded detection band marks 95-100%, not a pass threshold.

2. Board-Level Pulse / Duty Sweep



At the standard 260-DAC-sample period, a 12-sample pulse occupies only 4.6% of the waveform and still gives 99.4% validity. The 24- and 64-sample cases give 98.0% and 99.4%. Doubling the period to 520 samples reduces validity to 75.3% (24-sample pulse) and 88.2% (64-sample pulse), despite similar tone RMSE. Impulses can be narrow and low-duty, but repetition should not be made arbitrarily sparse.

Figure 2. Duty is derived as pulse length / repetition period. The p520_e16t32 run has no calibration_performance.csv; no final-skew or Stage 5 value is assigned.

3. Implication for Selling Point 1

Evidence supports	Not yet demonstrated
Sparse impulse generation; reliable recovery through the existing DAC-to-ADC chain; low temporal occupancy; a measured strength/disturbance trade-off.	Independent passive coupling into an otherwise unmodified mission-signal path. A later coupling demo remains a publication-level option.

4. Status of Selling Point 2

Event detection is strong. Offset observability remains limited; dither gain has no stable calibration factor; fine-skew is unreliable under measured dispersion; bandwidth calibration was not pursued. Tone-based calibration remains the stable main path. No three-parameter separation is claimed.

5. Decisions for the September Meeting

- 1. Are the current board results sufficient to begin consolidating the work toward publication?
- 2. Should the next hardware step focus on an independent passive-coupling demonstration to strengthen selling point 1?
- 3. What scope and division of work should be assigned to the additional FPGA student?

Data note. All ten summary rows were independently recomputed from committed CSV captures. Stage 5 exists for nine runs; its per-run calibrated parallel-average means span 33.90-45.58 dB SNDR, 32.80-53.96 dB SFDR, and 5.34-7.28 bits ENOB. These ranges are reported for provenance, not as evidence of dither-based calibration.